AD-CISPO: Saliency-Budgeted Clipped Importance Sampling for Reasoning RL

Abstract

Long-chain reasoning reinforcement learning depends on stable policy-gradient updates over token sequences, but current clipped objectives usually apply the same clipping radius to every generated token. This proposal studies AD-CISPO, a token-adaptive extension of MiniMax's clipped importance-sampling policy optimization (CISPO). The central hypothesis is that CISPO's dense token-gradient property can be preserved while improving sample efficiency by allocating a wider upper importance-sampling corridor to tokens that a frozen reference model treats as structurally important for subsequent generated tokens. The proposed novelty is a mean-preserving, saliency-budgeted upper clipping rule: the average clipping budget remains equal to the scalar CISPO baseline, but the budget is redistributed across generated tokens using exact future-attention in-degree from selected reference-model layers.

1 Problem and Motivation

Reinforcement learning has become a practical way to elicit mathematical and long-form reasoning from language models, especially when correctness can be checked by rule-based or programmatic rewards. PPO-style RLHF established clipped policy-gradient training as a scalable alignment method [1, 2], while GRPO removed the critic by using group-relative advantages and has been widely used for mathematical reasoning [3, 4]. More recent systems such as DAPO and MiniMax-M1 focus on stabilizing long-chain reasoning RL at scale [5, 6].

The unresolved problem is that scalar clipping treats every generated token as equally deserving of update freedom. In chain-of-thought-style reasoning, this is unlikely to be true. A token that introduces an intermediate quantity, selects an operation, or anchors a later conclusion may matter more than a formatting token or locally redundant phrase. A single global clipping radius therefore creates an undesirable tradeoff:

- if the radius is tight, important reasoning tokens may be prevented from moving enough when the reward signal identifies a better trajectory;
- if the radius is loose, low-value or unstable tokens receive the same update corridor, increasing variance and drift;
- if PPO/GRPO surrogate clipping suppresses token updates, the model can lose useful gradient signal exactly when the sampled policy has moved outside the trust corridor.

CISPO directly targets the third issue by clipping the importance-sampling (IS) weight itself and using the detached clipped weight to scale $\log \pi_\theta$, so clipped tokens still contribute gradients [6]. AD-CISPO asks whether the remaining scalar-budget issue can be addressed without abandoning CISPO's dense-gradient design.

2 Existing Solutions and Limitations

PPO and GRPO. PPO clips the policy-ratio surrogate to control policy updates while allowing multiple optimization epochs per batch [1]. GRPO adapts this family of methods to reasoning models by computing advantages from a group of sampled responses rather than a learned critic [3]. These methods are simple and effective, but their token-level clipping rules still use fixed trust-region hyperparameters.

DAPO and asymmetric clipping. DAPO improves long-chain reasoning RL with techniques such as a higher upper clip, dynamic sampling, token-level policy-gradient loss, and overlong reward shaping [5]. This is an important baseline because it shows that the upper clipping corridor matters for reasoning RL. However, DAPO still treats the upper corridor as a sequence-wide or globally configured scalar, not as a budget to be distributed across tokens.

CISPO. MiniMax-M1 proposes CISPO, which clips IS weights rather than clipping away token updates and reports improved RL efficiency in large-scale reasoning training [6]. CISPO is the strongest starting point for this proposal because it preserves per-token gradient density. Its limitation for this project is that the upper IS clipping radius remains scalar.

Attention-based saliency. Transformers route information through attention mechanisms [7], and attention patterns can provide useful structural signals. However, attention weights should not be treated as direct causal explanations of model behavior [8]. Attention sinks are a concrete example: initial or special tokens may receive high attention for non-semantic reasons [9]. AD-CISPO therefore uses frozen-reference attention only as an update-budget heuristic, not as a claim about human-interpretable explanation, and explicitly masks prompt and sink tokens.

3 Research Gap

Existing reasoning-RL objectives answer two separate questions only partially:

1. How can policy-gradient updates remain stable under off-policy drift? PPO, GRPO, DAPO, and CISPO address this through clipped ratios or clipped IS weights.

2. Which generated tokens deserve a larger or smaller update corridor? Existing methods largely do not answer this; they apply scalar clipping budgets across all tokens.

The gap is a token-budgeted trust-region objective for language-model RL: one that keeps the same average clipping budget as a scalar baseline, but reallocates that budget according to a frozen, reproducible structural signal. AD-CISPO is designed to fill this gap for CISPO-style clipped IS-weight policy gradients.

4 Proposed Solution

AD-CISPO replaces the scalar upper IS clipping radius in CISPO with a token-adaptive upper radius. The lower bound remains inactive by default to match the MiniMax-style CISPO baseline. Each generated token receives a multiplier computed from frozen-reference saliency, and the multipliers are normalized so their mean over valid generated tokens is 1. This preserves the same average clipping budget as scalar CISPO while changing where the budget is spent.

Core hypothesis. For reasoning tasks with sparse sequence-level rewards, reallocating CISPO's upper clipping budget toward structurally reused generated tokens will improve learning efficiency or stability relative to scalar CISPO, while shuffled or uniform multipliers should remove the gain if the saliency signal is doing real work.

What is novel. The proposal's novelty is not the use of attention as a saliency heuristic by itself. The novel contribution is the combination of:

1. CISPO-compatible token-adaptive upper IS clipping that preserves dense token gradients;
2. mean-preserving multiplier normalization, so AD-CISPO changes budget allocation rather than increasing the total average clipping budget;
3. exact frozen-reference future-attention in-degree, computed only over valid generated tokens and excluding prompt/sink tokens;
4. blockwise implementation that avoids materializing full attention tensors while preserving exact selected-layer attention weights;
5. falsifiable ablations that separate saliency from extra variance, asymmetric clipping, and reference-model overhead.

5 Notation

Symbol	Meaning	Shape or domain
q	prompt/question	text
i	index of a sampled response within a prompt group	$i \in \{1, \ldots, G\}$
o_i	response i in a sampled group	token sequence
t	token position in response i whose saliency, IS weight, or multiplier is being computed	$t \in M_i$
j	later token position in response i that attends back to position t	$j \in M_i$, $j > t$
u	summation index over valid generated positions in response i	$u \in M_i$
$o_{i,t}$	token at position t of response i	token id
G	number of responses sampled per prompt	positive integer
π_θ	current policy being optimized	language model
$\pi_{\theta_{\text{old}}}$	rollout policy that produced the batch	language model
π_{ref}	frozen reference model for saliency and optional KL	language model
$r_{i,t}(\theta)$	IS weight $\frac{\pi_\theta(o_{i,t}\mid q,o_{i,<t})}{\pi_{\theta_{\text{old}}}(o_{i,t}\mid q,o_{i,<t})}$	positive scalar
$\hat{A}_i$	group-relative sequence advantage for response i	scalar
M_i	valid generated-token mask for response i	token-position set
$S_{i,t}$	raw frozen-reference saliency for token $o_{i,t}$	nonnegative scalar
$m_{i,t}$	normalized token multiplier	nonnegative scalar, mean 1 over M_i
$\epsilon^{IS}_{\text{low}}$	lower IS clipping radius	nonnegative scalar
$\epsilon^{IS}_{\text{high}}$	scalar upper IS clipping radius	positive scalar
$\epsilon^{IS}_{i,t,\text{high}}$	token-adaptive upper IS clipping radius	positive scalar

Index conventions. For each prompt, the model samples G responses; i selects one of them. Within response i, positions t, j, and u all refer to token indices in the same sequence, but play different roles: t is the position whose saliency or clipping budget is being assigned; j is a *later* position whose attention query looks back at t; and u is a dummy index used only when averaging or summing over all valid generated positions in M_i.

6 Baseline: MiniMax-Style CISPO

The CISPO baseline starts from the off-policy REINFORCE objective with the IS weight treated as a stop-gradient coefficient:

$$J_{\text{ISPG}}(\theta) = \mathbb{E}\left[\frac{1}{\sum_{i=1}^{G}|o_i|}\sum_{i=1}^{G}\sum_{t=1}^{|o_i|}\text{sg}(r_{i,t}(\theta))\,\hat{A}_i\log\pi_\theta(o_{i,t}\mid q, o_{i,<t})\right]. \tag{1}$$

CISPO clips the IS weight itself:

$$\hat{r}_{i,t}(\theta) = \text{clip}\left(r_{i,t}(\theta), 1-\epsilon_{\text{low}}^{IS}, 1+\epsilon_{\text{high}}^{IS}\right), \tag{2}$$

and optimizes

$$J_{\text{CISPO}}(\theta) = \mathbb{E}\left[\frac{1}{\sum_{i=1}^{G}|o_i|}\sum_{i=1}^{G}\sum_{t=1}^{|o_i|}\text{sg}(\hat{r}_{i,t}(\theta))\,\hat{A}_i\log\pi_\theta(o_{i,t}\mid q, o_{i,<t})\right]. \tag{3}$$

A faithful baseline for this proposal sets the KL coefficient to zero and avoids an active lower IS-weight bound, following the MiniMax-M1 description of CISPO [6]. This isolates the effect of the upper clipping budget.

7 AD-CISPO Objective

AD-CISPO replaces the scalar upper clipping radius with a token-adaptive upper radius:

$$\epsilon_{i,t,\text{high}}^{IS} = \epsilon_{\text{high}}^{IS} m_{i,t}. \tag{4}$$

The clipped weight becomes

$$\hat{r}_{i,t}^{\text{AD}}(\theta) = \text{clip}\left(r_{i,t}(\theta), 1-\epsilon_{\text{low}}^{IS}, 1+\epsilon_{i,t,\text{high}}^{IS}\right), \tag{5}$$

and the objective is

$$J_{\text{AD-CISPO}}(\theta) = \mathbb{E}\left[\frac{1}{\sum_{i=1}^{G}|o_i|}\sum_{i=1}^{G}\sum_{t=1}^{|o_i|}\text{sg}\big(\hat{r}_{i,t}^{\text{AD}}(\theta)\big)\,\hat{A}_i\log\pi_\theta(o_{i,t}\mid q, o_{i,<t})\right]. \tag{6}$$

The adaptive multiplier is computed from frozen-reference saliency. Here u ranges over all valid generated positions in response i to form the mean saliency in the denominator, while t is the position whose multiplier is being computed:

$$\tilde{m}_{i,t} = \frac{S_{i,t}}{\frac{1}{|M_i|}\sum_{u\in M_i} S_{i,u}}. \tag{7}$$

If all valid saliency values are zero or non-finite, $\tilde{m}_{i,t}$ falls back to 1 for valid tokens. If lower or upper multiplier bounds are used, they must be followed by renormalization or projection so that

$$\frac{1}{|M_i|} \sum_{t \in M_i} m_{i,t} = 1. \tag{8}$$

A plain clamp is insufficient because it changes the total clipping budget.

8 Saliency Estimation

The main AD-CISPO implementation uses exact frozen-reference future-attention in-degree. For a selected reference layer ℓ and head h, let $a_{i,j,t}^{(\ell,h)}$ denote the causal attention probability from the query at token position j to the key/value position t. Position t is the saliency target; position j is each later valid generated token that attends to t. The saliency assigned to token $o_{i,t}$ is

$$S_{i,t} = \frac{1}{|L_{\text{top}}|} \sum_{\ell \in L_{\text{top}}} \frac{1}{H_\ell} \sum_{h=1}^{H_\ell} \sum_{\substack{j \in M_i \\ j > t}} a_{i,j,t}^{(\ell,h)}. \tag{9}$$

Thus a token is salient when later valid generated tokens attend to it under the frozen reference model. The strict condition $j > t$ prevents self-attention from inflating saliency. M_i excludes prompt positions and special-token sinks from both source queries and target keys.

The implementation computes this quantity blockwise from reference-model post-RoPE query and key states. This avoids materializing a full batch of $T \times T$ attention matrices while preserving exact softmax attention weights for the selected layers, including causal, padding, and sliding-window masks.

A cheaper ablation is frozen-reference key-vector norm:

$$S_{i,t}^{\text{KV}} = \frac{1}{|L_{\text{top}}|} \sum_{\ell \in L_{\text{top}}} \left\| k_{i,t}^{(\ell)} \right\|_2. \tag{10}$$

It should be reported as a proxy baseline, not as the primary saliency definition.

9 Implementation Plan

The minimum implementation should use the existing Curious training path and GSM8K reward setup [10]. The intended experimental model is a small reasoning-capable open model, such as the repository's Qwen3-1.7B configuration, so that full ablations are affordable on a single H100.

Implementation flow: $\pi_{\theta_{\text{old}}}$ samples grouped responses $\rightarrow$ rule-based verifier computes rewards and group advantages $\rightarrow$ frozen π_{ref} computes log-probs and saliency $\rightarrow$ saliency is normalized into $m_{i,t}$ $\rightarrow$ token thresholds $\epsilon_{i,t,\text{high}}^{IS} = \epsilon_{\text{high}}^{IS} m_{i,t}$ $\rightarrow$ CISPO loss uses detached clipped IS weights $\rightarrow$ metrics log reward, exact match, clipping, saliency, and runtime overhead.

Implementation contract. A mathematically aligned AD-CISPO implementation should satisfy the following:

1. CISPO loss must use a detached clipped IS weight multiplying $\log \pi_\theta$, not a PPO/GRPO surrogate minimum.

2. AD-CISPO must be enabled only with the CISPO loss, unless explicitly naming and evaluating an adaptive-GRPO ablation.
3. MiniMax-style CISPO should set KL weight to zero and avoid an active lower IS-weight bound.
4. Token-adaptive clipping should change only the upper IS clipping radius unless a lower-bound ablation is deliberately introduced.
5. Saliency tensors should align with the generated token whose IS weight is being clipped. In next-token prediction implementations, generated-token saliency must align with next-token log-probability terms, not the preceding hidden-state position.
6. Claims about compute and memory must distinguish active-policy overhead from frozen-reference overhead.

10 Evaluation Plan

Datasets and tasks. The evaluation should use a staged math-reasoning suite rather than a single benchmark:

1. **GSM8K** should remain the first target, because it has a clear exact-answer reward and is already supported in Curious [10]. It is the right dataset for fast debugging, seed sweeps, and early reward-curve comparisons.
2. **SVAMP** should be used as an evaluation-only robustness check for arithmetic word-problem variations [11]. This helps test whether AD-CISPO improves reasoning robustness rather than merely fitting GSM8K surface patterns.
3. **MATH and MATH-500** should be the next difficulty tier. MATH provides competition-style problems with step-by-step solutions [12], while MATH-500 is a smaller representative subset commonly used for cheaper repeated evaluation [13]. These tasks test whether the token-adaptive clipping signal transfers to longer derivations and more diverse mathematical domains.
4. **AIME/AMC-style held-out sets** should be reserved for final stress testing once answer extraction is stable. If broader olympiad-level coverage is needed, add the mathematics split of OlympiadBench as an optional evaluation-only benchmark [14].

Training should begin with GSM8K and expand only after scalar CISPO and AD-CISPO have matched data loading, answer extraction, rollout count, and compute budgets. Harder datasets should be used primarily to measure transfer, not to tune the AD-CISPO saliency heuristic.

Baselines. The minimum comparison set is:

1. GRPO/PPO-style surrogate baseline.
2. DAPO-style asymmetric clipping or clip-high baseline.
3. MiniMax-style CISPO: no KL, no active lower IS bound, scalar upper IS radius.
4. AD-CISPO with exact future-attention in-degree saliency.
5. AD-CISPO with KV-norm proxy saliency.
6. Shuffled-saliency and uniform-multiplier controls.
7. Optional adaptive-GRPO ablation, clearly separated from AD-CISPO.

Metrics. Report both task performance and optimizer behavior:

- exact-match accuracy, verifier reward, and pass@k by dataset and difficulty tier;
- reward-per-token and reward-per-GPU-hour;
- upper-clip saturation rate and token update coverage;
- entropy, approximate KL to the reference model when logged, and response length;
- AD-CISPO multiplier mean, min, max, saliency mean, and saliency concentration;
- frozen-reference overhead in wall-clock time and peak memory.

Decision rule. AD-CISPO should be considered successful only if it improves exact-match or reward efficiency over scalar CISPO at comparable compute, or if it provides a clear stability advantage such as lower collapse rate or smoother reward improvement. A gain that also appears for shuffled saliency should be treated as evidence for variance injection or changed clipping distribution, not for the proposed saliency mechanism.

11 Milestones

1. **Week 1:** verify MiniMax-style CISPO against GRPO on GSM8K with matched model, batch size, rollout count, and reward function.
2. **Week 2:** profile exact future-attention saliency and KV-norm saliency on short runs; confirm mean-preserving multiplier statistics.
3. **Week 3:** run AD-CISPO, scalar CISPO, KV-norm, shuffled, and uniform ablations with matched seeds.
4. **Week 4:** analyze reward curves, clipping dynamics, saliency concentration, runtime overhead, and representative trajectory changes.

12 Risks, Ablations, and Fallbacks

Risk: attention saliency is not causal. Attention weights may not explain model behavior [8]. This is why AD-CISPO is framed as a budget-allocation heuristic. The shuffled saliency ablation is required; gradient-based or leave-one-token-out saliency can be explored if attention in-degree fails.

Risk: attention sinks distort saliency. Initial and special tokens can attract attention for non-semantic reasons [9]. The implementation must mask prompt and sink tokens, use generated-token-only normalization, and report saliency concentration diagnostics.

Risk: frozen-reference overhead dominates. Exact future-attention in-degree requires additional reference-model work. The fallback is KV-norm saliency, fewer top layers, larger attention blocks, or saliency recomputation at a lower cadence.

Risk: budget concentration increases variance. Large multipliers may overemphasize a small set of tokens. The fallback is a bounded multiplier projection with post-clamp renormalization, plus reporting clip-min, clip-max, and saturation metrics.

13 Expected Impact

If successful, AD-CISPO would provide evidence that reasoning-RL trust regions should be token-budgeted rather than uniformly scalar. The contribution would be useful even if the exact attention heuristic is later replaced: the broader idea is to preserve the average IS clipping budget while assigning update freedom where a frozen model predicts later tokens will reuse earlier generated content. A negative result would still be informative because it would test whether CISPO's scalar clipping is already sufficient, and whether attention-derived structural signals are too weak or too noisy for optimizer control.

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